

SBT-DF204 — Computer Forensics Case Study

Investigating Harassment Email Traffic With Wireshark

Trainee Name: Oluyede Tobi Steven

Registration Number: 202513549PF

Course: SBT-DF204 — Computer Forensics Case Study

Date: 05 October 2026

Instructor: Aminu Idris, AMCPN

Environment: Isolated Virtual Laboratory

Executive Summary

This investigation examined the `nitroba.pcap` network capture to identify the device and browser activity associated with a simulated harassment email. Wireshark analysis identified client IP `192.168.15.4` and MAC `00:17:f2:e2:c0:ce`. Frame 83601 showed the client submitting a harassment message through `www.willselfdestruct.com` to `lilytuckrige@yahoo.com`. Related Gmail traffic contained the identifier `jcoachj@gmail.com`, which is consistent with the Chemistry 109 roster entry for Johnny Coach. The evidence supports an association with the device and account, but does not conclusively prove who physically operated the device.

Objectives

1. Verify the integrity of the PCAP.
2. Identify the client IP and MAC addresses.
3. Identify the harassment web service and relevant HTTP traffic.
4. Examine the harassment submission and related Gmail activity.
5. Compare recovered account information with the class roster.
6. Establish the event timeline and assess attribution.

Tools Used

- Wireshark — packet and HTTP analysis.
- Capinfos — PCAP metadata.

- SHA-256/SHA-1/MD5 — evidence integrity verification.
- cmp — working-copy verification.
- Kali Linux terminal — evidence management and forensic commands.
- Supplied Nitroba case materials and PCAP.

Methodology

The original PCAP was preserved and a working copy was created and hash-verified. Wireshark was then used to identify the relevant client, HTTP connections, TCP streams, harassment submission, and Gmail session information. Packet timestamps were used to construct the timeline. The recovered Gmail identifier was compared with the supplied Chemistry 109 roster. Findings were separated into direct observations and reasonable inferences, with attribution limitations documented.

Item	Detail
Original filename	nitroba.pcap
Source	Digital Corpora Nitroba University Harassment Scenario
Working copy	working/nitroba_working.pcap
Original size	56,180,821 bytes
SHA-256	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA-1	65656392412add15f93f8585197a8998aaeb50a1
MD5	9981827f11968773ff815e39f5458ec8

Packets 94,410

Capture start 2008-07-21 21:51:07.095278

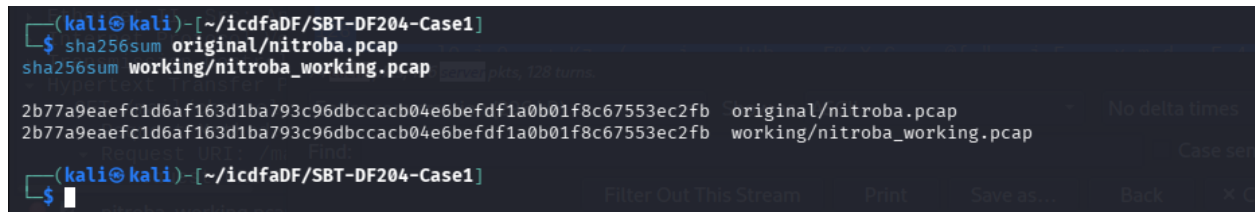
Capture end 2008-07-22 02:13:47.046029

Command used:

```
cd ~/icdfaDF/GBT-DF204-Case1
```

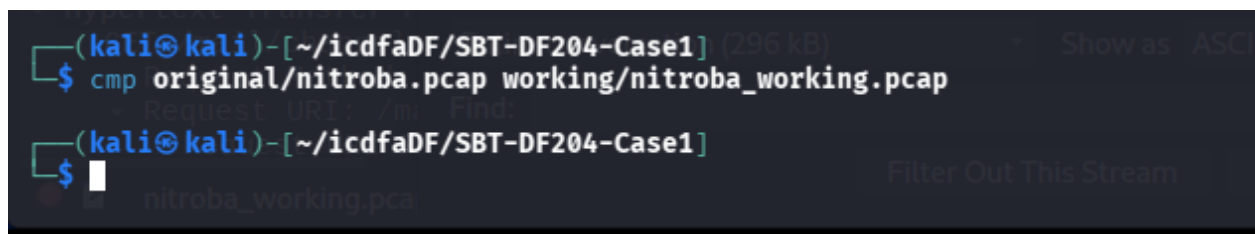
```
sha256sum original/nitroba.pcap
```

```
sha256sum working/nitroba_working.pcap
```

A terminal window with a dark background. The prompt is (kali@kali)-[~/icdfaDF/GBT-DF204-Case1]. The user enters sha256sum original/nitroba.pcap. The output shows two identical SHA256 hashes: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb. The first hash is for original/nitroba.pcap and the second is for working/nitroba_working.pcap. The terminal also shows some faint background text like "No delta times" and "Case set".

```
(kali@kali)-[~/icdfaDF/GBT-DF204-Case1]  
$ sha256sum original/nitroba.pcap  
sha256sum working/nitroba_working.pcap  
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb original/nitroba.pcap  
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap  
(kali@kali)-[~/icdfaDF/GBT-DF204-Case1]  
$
```

```
cmp original/nitroba.pcap working/nitroba_working.pcap
```

A terminal window with a dark background. The prompt is (kali@kali)-[~/icdfaDF/GBT-DF204-Case1]. The user enters cmp original/nitroba.pcap working/nitroba_working.pcap. The output shows the file sizes: (296 kB) for original/nitroba.pcap and (296 kB) for working/nitroba_working.pcap. The terminal also shows some faint background text like "Show as ASCII" and "Filter Out This Stream".

```
(kali@kali)-[~/icdfaDF/GBT-DF204-Case1] (296 kB)  
$ cmp original/nitroba.pcap working/nitroba_working.pcap  
(kali@kali)-[~/icdfaDF/GBT-DF204-Case1]  
$
```

```
capinfos working/nitroba_working.pcap
```

```
(kali㉿kali)-[~/icdfadF/GBT-DF204-Case1]
$ cmp original/nitroba.pcap working/nitroba_working.pcap

(kali㉿kali)-[~/icdfadF/GBT-DF204-Case1]
$ capinfos working/nitroba_working.pcap
File name:      working/nitroba_working.pcap
File type:      Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size:      56 MB
Data size:      54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eae1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410
```

The original and working-copy SHA-256 values matched, and cmp produced no output, indicating that the working copy is byte-for-byte identical to the original.

E02 — Contact with willselfdestruct.com

Frame 82936

Timestamp:

22 Jul 2008 02:03:43.825871 EDT

Network information:

Source MAC: 00:17:f2:e2:c0:ce

Source IP: 192.168.15.4

Destination IP: 69.25.94.22

Destination port: 80

TCP stream: 1680

HTTP request:

GET /secure/submit HTTP/1.1

Host: www.willselfdestruct.com

User-Agent:

Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)

Wireshark filter

frame.number == 82936

or:

ip.src == 192.168.15.4 && http.host == "www.willselfdestruct.com"

```

Frame 82936: Packet, 596 bytes on wire (4768 bits), 596 bytes captured (4768 bits)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
Transmission Control Protocol, Src Port: 35984, Dst Port: 80, Seq: 1, Ack: 1, Len: 526
Hypertext Transfer Protocol
  GET /secure/submit HTTP/1.1\r\n
    Request Method: GET
    Request URI: /secure/submit
    Request Version: HTTP/1.1
    Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*\r\n
    Referer: http://email.about.com/gi/dynamic/offsite.htm?zi=1/XJ&sdn=email&cdn=compute&tm=17&gps=101_1829_7
    Accept-Language: en-us\r\n
    Accept-Encoding: gzip, deflate\r\n
    User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
    Host: www.willselfdestruct.com\r\n
    Connection: Keep-Alive\r\n
  \r\n
  [Response in frame: 82998]
  [Full request URI: http://www.willselfdestruct.com/secure/submit]
```


HTTP:

POST /secure/submit HTTP/1.1

Host: www.willselfdestruct.com

Content-Type: application/x-www-form-urlencoded

Content-Length: 188

The form data identifies:

to = lilytuckrige@yahoo.com

subject = you can't find us

message = [harassing message]

For the final report, **do not reproduce the threatening message unnecessarily**. Identify it as the captured harassment message.

Wireshark filter

frame.number == 83601

or:

ip.src == 192.168.15.4 && http.request.method == "POST"

```

HyperText Transfer Protocol
  POST /secure/submit HTTP/1.1\r\n
    Request Method: POST
    Request URI: /secure/submit
    Request Version: HTTP/1.1
    Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*\r\n
    Referer: http://www.willselfdestruct.com/secure/submit\r\n
    Accept-Language: en-us\r\n
    Content-Type: application/x-www-form-urlencoded\r\n
    Accept-Encoding: gzip, deflate\r\n
    User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
    Host: www.willselfdestruct.com\r\n
  Content-Length: 188\r\n
    [Content length: 188]
    Connection: Keep-Alive\r\n
    Cache-Control: no-cache\r\n
    \r\n
    [Response in frame: 83604]
    [Full request URI: http://www.willselfdestruct.com/secure/submit]
    File Data: 188 bytes
  HTML Form URL Encoded: application/x-www-form-urlencoded
  Form item: "to" = "lilytuckrige@yahoo.com"
    Key: to
    Value: lilytuckrige@yahoo.com
  Form item: "from" = ""
  Form item: "subject" = "you can't find us"
  Form item: "message" = "and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running. "
  Form item: "type" = "0"
  Form item: "ttl" = "30"
  Form item: "submit.x" = "92"
  Form item: "submit.y" = "26"

```

E04 — Gmail browser/session evidence

Frame 83326

Frame 83326 occurred during the same investigation period.

Network:

Source MAC: 00:17:f2:e2:c0:ce

Source IP: 192.168.15.4

Destination IP: 74.125.19.17

Destination port: 80

TCP stream: 1611

HTTP:

POST /mail/channel/bind

Host: mail.google.com

Browser:

Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)

Most importantly, the HTTP cookie contains:

gmailchat=jcoachj@gmail.com/475090

Wireshark filter

frame.number == 83326

or:

ip.src == 192.168.15.4 && http.host == "mail.google.com"

```
Accept: */*\r\n
Accept-Language: en-us\r\n
Referer: http://mail.google.com/mail/?ui=1&view=page&name=js&ver=167ge8cpe09rv\r\n
Content-Type: application/x-www-form-urlencoded\r\n
Accept-Encoding: gzip, deflate\r\n
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
Host: mail.google.com\r\n
Content-Length: 51\r\n
[Content length: 51]
Connection: Keep-Alive\r\n
Cache-Control: no-cache\r\n
Cookie [...]: GX=DQAAAG8AAAAAm2ow8LqM60qoQ5w2jVJ-zHIfuyAQ3GUkvcv4N9vQ6lWuLpVMCmw1Jhmlm9_P3qZbyTWkIWDo5cnuJl
Cookie pair: GX=DQAAAG8AAAAAm2ow8LqM60qoQ5w2jVJ-zHIfuyAQ3GUkvcv4N9vQ6lWuLpVMCmw1Jhmlm9_P3qZbyTWkIWDo5c
Cookie pair: S=gmail=L5hb7hHJ9B97n6StWA4FvA:gmail_yj=-0oenmU7qTeuQ1dsN3B1kg:gmpoxy=6uatNcZZmB8:gmprox
Cookie pair: GMAIL_AT=xn3j32oktf2a0q6oa3k9sfr6d09yzf
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: PREF=ID=8fc081df5e738a3c:TM=1210743469:LM=1216706486:GM=1:S=vvxeHX0oIXNyR8Zj
Cookie pair: NID=13=tJ7LtEc6z12iH4BP_IPyV0gGhi4aLcZoJcJAF7l-9JQ2AeoD8owG9Njt0p7T5tuskknNgEKMRAN9P49vI4E
Cookie pair: __utmx=173272373.00000983192309928271:2:
Cookie pair: __utmxx=173272373.00000983192309928271:1216706401:2592000
Cookie pair: SID=DQAAAGwAAACH8Y_j5izp1fdbDJzwdRFDgtU3aaeZKwgZ7DwUjYpLoqH7F1_E-X5taC4l0uvzXtrVeE6Zq1gco
Cookie pair: TZ=-60
Cookie pair: GMAIL_HELP=hosted:0
\r\n
[Response in frame: 83328]
[Full request URI: http://mail.google.com/mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf&ui=1&VER=20]
File Data: 51 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
  Form item: "count" = "1"
  Form item: "req0_type" = "i"
  Form item: "req0_time" = "172110"
  Form item: "req0_evtype" = "7"
```

Finding: The captured client was communicating with Gmail and the browser traffic contained an account identifier of jcoachj@gmail.com.

6. E05 — Confirmation of Gmail session

Frame 84201

Arrival Time:

22 Jul 2008 02:04:55.380722 EDT

Network:

Source MAC: 00:17:f2:e2:c0:ce

Source IP: 192.168.15.4

Destination IP: 74.125.19.17

TCP stream: 1611

HTTP:

GET /mail/channel/bind?...TYPE=terminate...

Host: mail.google.com

The same session contains:

SID=67F8DC1634D9D313

and:

gmailchat=jcoachj@gmail.com/475090

Wireshark filter

frame.number == 84201

```

▼ Frame 84201: Packet, 1366 bytes on wire (10928 bits), 1366 bytes captured (10928 bits)
  Encapsulation type: Ethernet (1)
  Arrival Time: Jul 22, 2008 02:04:55.380722000 EDT
  UTC Arrival Time: Jul 22, 2008 06:04:55.380722000 UTC
  Epoch Arrival Time: 1216706695.380722000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 646.172000 milliseconds]
  [Time delta from previous displayed frame: 646.172000 milliseconds]
  [Time since reference or first frame: 4 hours, 13 minutes, 48.285444000 seconds]
  Frame Number: 84201
  Frame Length: 1366 bytes (10928 bits)
  Capture Length: 1366 bytes (10928 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http || tcp.port == 80 || http2]
▼ Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
  ▸ Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
  ▸ Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
  Type: IPv4 (0x0800)
  [Stream index: 19]
  Frame check sequence: 0x020dd0dc [unverified]
  [FCS Status: Unverified]
▼ Internet Protocol Version 4, Src: 192.168.15.4, Dst: 74.125.19.17
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 1348
  Identification: 0x1ecb (7883)
  ▸ 010. .... = Flags: 0x2, Don't fragment

```

```

  ▸ [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (1296 bytes)
▼ Hypertext Transfer Protocol
  ▸ GET /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf&ui=1&SID=67F8DC1634D9D313&RID=42256&TYPE=termini
  Request Method: GET
  ▸ Request URI: /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf&ui=1&SID=67F8DC1634D9D313&RID=42256&
  Request URI Path: /mail/channel/bind
  ▸ Request URI Query: at=xn3j32oktf2a0q6oa3k9sfr6d09yzf&ui=1&SID=67F8DC1634D9D313&RID=42256&TYPE=termini
  Request Version: HTTP/1.1
  Accept: */*\r\n
  Referer: http://mail.google.com/mail/?ui=1&view=page&name=js&ver=167ge8cpe09rv\r\n
  Accept-Language: en-us\r\n
  Accept-Encoding: gzip, deflate\r\n
  User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
  Host: mail.google.com\r\n
  Connection: Keep-Alive\r\n
▼ Cookie [...]
  Cookie pair: GX=DQAAAG8AAAAm2ow8LqM60qoQ5w2jVJ-zHIfuyAQ3GUKvcv4N9vQ6lwLpVMcmw1Jhmlm9_P3qZbyTwkIWDo5cnuJI
  Cookie pair: S=gmail=L5hb7hHJ9B97n6StWA4FvA:gmail_yj=-0oenmU7qTeuQ1dsN3B1kg:gmpoxy=6uatNcZZmB8:gmprox
  Cookie pair: GMAIL_AT=xn3j32oktf2a0q6oa3k9sfr6d09yzf
  Cookie pair: gmailchat=jcoachj@gmail.com/475090
  Cookie pair: PREF=ID=8fc081df5e738a3c:TM=1210743469:LM=1216706486:GM=1:S=vvxeHX0oIXNyR8Zj
  Cookie pair: NID=13=tJ7LtEc6z12iH4BP_IPyV0gGhi4aLcZoJcJf7l-9JQ2AeoD8oWG9NjtOp7T5tuskNgEKMRAN9P49vI4E
  Cookie pair: __utmx=173272373.00000983192309928271:2:
  Cookie pair: __utmx=173272373.00000983192309928271:1216706401:2592000
  Cookie pair: SID=DQAAAGwAAACH8Y_j5izp1fdbDJzwdRFDGtU3aaeZKwGz7DwUjYpLoqH7F1_E-X5taC4l0uvzXtrVeE6Zq1gco
  Cookie pair: TZ=-60
  Cookie pair: GMAIL_HELP=hosted:0
\r\n
[Full request URI: http://mail.google.com/mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf&ui=1&SID=6

```

Finding: Frame 84201 provides a second observation of the same Gmail identifier and session context approximately 31 seconds after the harassment POST.

E06 — Chemistry 109 roster

The supplied case materials contain the Chemistry 109 class list.

Relevant entry:

Johnny Coach

The captured Gmail identifier:

jcoachj@gmail.com

is reasonably consistent with the name **Johnny Coach**.

However, this is an **inference**, not a direct statement in the roster that Johnny Coach owns the Gmail address.

```
(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case1]
$ pdftotext -layout evidence/nitroba_case_materials.pdf evidence/nitroba_case_materials.txt
grep -n -A30 -B5 "Chemistry 109 class list" evidence/nitroba_case_materials.txt
164-5223.531 Request Method: GET
165-5223.531 Request URI: /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf&ui=1&SID=67F
166-5223.593 Request URI: /mail/channel/bind
167-5223.593 State University
167-So who did it?
168-5223.681 Request Version: HTTP/1.1
169-5223.681 http://mail.google.com/mail/?ui=1&view=page&name=js&ver=167g
170-5223.683 -Language: en-us\r\n
171-5223.683 -Encoding: gzip, deflate\r\n
172-5223.683 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
173-5223.683 Host: mail.google.com\r\n
174-5223.683 Connection: Keep-Alive\r\n
175-5223.683 Cookie [ ]: GX=DQAAAG8AAAAAAM2ow8LqM60qoQ5w2jVJ-zHIfuyAQ3Gukvcv4N9vQ6L
176-5223.683 Cookie pair: GX=DQAAAG8AAAAAAM2ow8LqM60qoQ5w2jVJ-zHIfuyAQ3Gukvcv4N9v
177-5223.683 Cookie pair: S=gmail=L5hb7hHJ9B97n6StWA4FvA:gmail_yj=-0oenmU7qTeuQ1
178-5223.683 Cookie pair: GMAIL_AT=xn3j32oktf2a0q6oa3k9sfr6d09yzf
179-5223.683 Cookie pair: gmailchat=jcoachj@gmail.com/475090
180-5223.683 Cookie pair: PREF=ID=8fc081df5e738a3c:TM=1216743469:LM=1216706486:6
181-5223.683 Cookie pair: NID=13=tJ7LEc6z12iH48P_IPyV0gGh14aLc26JcJAF7l-9JQ2Aeo
182-5223.683 Cookie pair: __utmx=173272373.000000983192309928271:2:
183-5223.683 Cookie pair: __utmx=173272373.000000983192309928271:1216706486:2592
184-5223.683 Cookie pair: SID=DQAAAGwAAACH8Y_j51zp1fddDJzwdRFDGtU3aae2KwgZ7DwUjY
185-5223.683 Cookie pair: TZ=-60
186-5223.683 Cookie pair: GMAIL_HELP=hosted:0
187-5223.683
188-5223.683
189-5223.683
190-5223.683
```

Timeline

Time (EDT)	Evidence	Event
------------	----------	-------

02:03:43.82587 1	Frame 82936	Client accesses <code>www.willselfdestruct.com</code>
02:04:24.311700	Frame 83601	Client submits harassment message to Lily's email address
02:04:55.38072 2	Frame 84201	Client communicates with Gmail; <code>jcoachj@gmail.com</code> identifier observed

The interval between the harassment POST and Frame 84201 is approximately:

31.069 seconds.

All three observations involve the client MAC:

`00:17:f2:e2:c0:ce`

and the harassment/Gmail traffic originates from:

`192.168.15.4`

Attribution assessment

Direct observations

The PCAP directly shows:

- Client IP `192.168.15.4`.
- Client MAC `00:17:f2:e2:c0:ce`.
- A connection to `www.willselfdestruct.com`.
- An HTTP POST submitting the harassment message.
- Gmail traffic from the same client.
- The browser cookie identifier `jcoachj@gmail.com`.
- The supplied Chemistry 109 roster contains **Johnny Coach**.

Reasonable inference

The jcoachj@gmail.com identifier is consistent with **Johnny Coach** in the supplied Chemistry 109 roster. The Gmail session and harassment submission are associated with the same captured client device.

Therefore, **Johnny Coach is a reasonable attribution candidate based on the available evidence.**

Limitation

The PCAP does **not** establish with certainty who physically operated the computer at the time.

The dorm room was shared by three women, and the scenario states that an open wireless router was installed. Consequently:

- An IP address cannot by itself identify a person.
- A MAC address identifies a network interface, not necessarily its user.
- A browser cookie associates activity with a browser/session, but does not independently prove who was physically operating the device.
- The jcoachj@gmail.com → Johnny Coach connection is based on corroboration/inference from the roster rather than an explicit roster statement.

Final conclusion

The forensic examination of the provided network capture identified a client device using IP address 192.168.15.4 and MAC address 00:17:f2:e2:c0:ce. At 02:03:43.825871 EDT, the client contacted www.willselfdestruct.com. At 02:04:24.311700 EDT, the same client submitted an HTTP POST to the service containing the victim's email address, lilytuckrige@yahoo.com, together with the subject and harassment message.

Related Gmail traffic from the same client contained the browser/session identifier gmailchat=jcoachj@gmail.com/475090. This identifier was observed in both Frames 83326 and 84201, with Frame 84201 occurring approximately 31 seconds after the harassment submission. The supplied Chemistry 109 roster contains a student named Johnny Coach, making the jcoachj@gmail.com identifier reasonably consistent with that student.

The evidence therefore provides a strong association between the harassment submission, the captured client device, and the Gmail browser/session identifier. However, the evidence does not independently establish who was physically operating the device at the time of transmission. The shared residence/network environment and the limitations of IP addresses, MAC addresses, and browser cookies must therefore be considered. Johnny Coach should

consequently be reported as a reasonable attribution candidate rather than as an identification established beyond doubt.